CHRIST (Deemed to be University), Bengaluru – 560 029

School of Engineering and Technology

MID SEMESTER EXAMINATION – September 2023

2360107

Programme Name: B.Tech.
Course Name: PHYSICS FOR ENGINEERS **Max. Marks: - 50**
Course Code: PH132P **Time: – 2 Hrs.**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / Course title & number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place in the hall.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer booklet. No sheets should be detached from the answer booklet.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer booklet.

Course Outcomes (COs): The students will able to

1. Explain the basic principles of quantum mechanics and its applications to simple systems.
2. Apply the principles of Electrical and Optical properties in different fields of Physics.
3. Analyze different properties of solids for various scientific applications.
4. Utilize the theoretical concepts to conduct experiments, to analyze and interpret data.

SECTION A (Answer ALL questions)

Q. No	Questions	Marks	CO	RBT
1	What are the properties of matter waves? Calculate the de-Broglie wavelength of Helium nucleus that is accelerated through 500 V.	10	1	3
2	State uncertainty principle. Write its mathematical form for different pair of variables. In a measurement that involved an uncertainty, the uncertainty in the speed of an electron was found to be 0.023 m/s. Calculate the uncertainty in momentum and the corresponding uncertainty involved in determining its position.	10	1	3
3	Describe the principle, construction and working of a semiconductor diode laser.	10	2	3
4	Explain the different types of optical fibers with necessary diagram. If an optical fiber has glass core of refractive index 1.5 and cladding of refractive index 1.48, calculate a) numerical aperture and b) angle of acceptance if the fiber is kept in air.	10	2	3

SECTION B (Answer any 1 question)

Q. No	Questions	Marks	CO	RBT
1	What is wave function? Give its physical significance and properties.	10	1	3
2	Discuss Eigen wave functions, probability densities and Normalization	10	1	3

Revised Bloom's Taxonomy (RBT) Levels :		
L1 – Remembering	L2 – Understanding	L3 – Applying
L4 – Analyzing	L5 – Evaluating	L6 - Creating

2360104
Register Number

CHRIST (Deemed to be University), Bangalore – 560 029

End Semester Examination –Oct/ Nov-2023

Bachelor of Technology (Physics Cycle) –I semester

PROGRAMME NAME: BTech **MAX. MARKS: 100**
COURSE NAME: Physics For Engineers **TIME: 3 hours**
COURSE CODE: PH132P

Course Outcomes (COs):

The students will able to:

1. Explain the basic principles of quantum mechanics and its applications to simple systems.
2. Apply the principles of Electrical and Optical properties in different fields of Physics.
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- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer booklet.
- Constants: $h=6.63 \times 10^{-34}$ Js; charge of electron $e=1.6 \times 10^{-19}$ C; speed of light $c=3 \times 10^{8}$ m/s; mass of electron, $m=9.1 \times 10^{-31}$ kg, Boltzaman constant $k_b=1.38 \times 10^{-23}$ J/K, 1 eV$=1.6 \times 10^{-19}$ J

Q. No	Questions	Marks	CO	RBT
1.	a) Calculate at what temperature a blackbody would emit the energy density per unit wavelength of 0.05 J/m^4 for a wavelength value of 7174 Å. **[10 Marks]** b) Explain qualitatively the double slit experiment with electrons with appropriate figures. **[10 Marks]**	20	1	L3
	OR			
2.	a) Define group velocity and derive an expression for group velocity based on the superposition of waves. **[10 Marks]** b) If an electron has a de Broglie wavelength of 2 nm, find its kinetic energy and group velocity, given that it has a rest mass energy of 511 keV. **[10 Marks]**	20	1	L3
3.	a) Derive time-independent one-dimensional Schrödinger's wave equation. **[10 Marks]** b) A quantum particle confined to a one-dimensional box of width 'a' is in its first excited state. What is the probability of finding the particle over an interval of (a/2) marked symmetrically at the center of the box. **[10 Marks]**	20	2	L3
	OR			
4.	a) Derive an expression for energy Eigenvalue and normalized wavefunction for a particle in a one-dimensional infinite potential	20		L3

	well. **[10 Marks]** b) An electron is bound in a one-dimensional infinite potential well of width 1 Å. Find its energy values in the ground state and first excited state. **[10 Marks]**			
5.	a) Write Fermi-Dirac distribution function. Explain how Fermi function varies with temperature. **[10 Marks]** b) Calculate the probability of an electron occupying an energy level 0.02 eV above the Fermi level at 400 K. **[10 Marks]**	20	3	L3
	OR			
6.	a) Define Conductors, Insulators and Semiconductors, in terms of bandgaps with appropriate figures. **[10 Marks]** b) The electron mobility and hole mobility of silicon are 0.17 m^2/volt-sec and 0.035 m^2/volt-sec respectively at room temperature. If the carrier density is known to be $1.1 \times 10^{16}/m^3$, calculate the resistivity of Silicon semiconductor. **[10 Marks]**	20	3	L3
7.	a) Define the bending moment of a beam and derive an expression for the bending moment of a rectangular beam. **[10 Marks]** b) A copper cube of each side 4cm is subjected to a tangential force of 9.98×10^4 N. The top face of the cube is sheared to 0.012 with respect to the bottom face. Find the shearing stress, shearing strain and the modulus of rigidity. **[10 Marks]**	20	4	L4
	OR			
8.	a) Explain the four types of polarization mechanisms involved in a dielectric material with diagrams. **[10 Marks]** b) Define polarization and polarizability. Find the polarization produced in a dielectric medium of relative permittivity 15 in presence of an electric field of 500 V/m. **[10 Marks]**	20	4	L4
9.	a) Describe the principle, construction and working of a semiconductor diode laser. (Construction and energy level diagram is a must) **[10 Marks]** b) The average output power of a laser source emitting laser beam of wavelength 6328 Å is 5 mW. Find the number of photon emitted per second by the laser source. **[10 Marks]**	20	5	L3
	OR			
10.	a) Explain the different types of optical fibers with the necessary diagrams. **[10 Marks]** b) If an optical fiber has glass core of refractive index 1.5 and cladding of refractive index 1.48, calculate a) numerical aperture and b) angle of acceptance if the fiber is kept in air. **[10 Marks]**	20	5	L3

Revised Bloom's Taxonomy (RBT) Levels:		
L1 - Remembering	L2 - Understanding	L3 - Applying
L4 - Analyzing	L5 - Evaluating	L6 - Creating

2360365
Register Number

CHRIST (Deemed to be University), Bangalore – 560 029

End Semester Examination –April 2024

Bachelor of Technology (Physics Cycle) – (II semester)

PROGRAMME NAME: BTech **MAX. MARKS: 100**

COURSE NAME: Physics for Engineers **TIME: 3 hours**

COURSE CODE: PH232P

Course Outcomes (COs):

The students will able to:

1. Explain the basic principles of quantum mechanics and its applications to simple systems.
2. Apply the principles of Electrical and Optical properties in different fields of Physics.
3. Analyze different properties of solids for various scientific applications.
4. Utilize the theoretical concepts to conduct experiments, to analyze and interpret data.

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Note: Constants: h=6.63 × 10^{-34} Js; Charge of electron e=1.6 x 10^{-19} C; Speed of light c=3 x 10^{8} m/s; Mass of electron, m=9.1 x 10^{-31} kg, Boltzmann constant k_b =1.38 x 10^{-23} J/K, 1 eV=1.6 x 10^{-19} J, Mass of neutron=1.675 x 10^{-27} kg , Mass of proton = 1.675 x 10^{-27} kg

Q. No	Questions	Marks	CO	RBT
1.	(a) Explain Planck's radiation law with formula. Calculate the energy density per unit wavelength in a blackbody cavity at a temperature 1500 K at a wavelength of 6000 Å. **[10 Marks]**	20	1	L3
	(b) Show that v_g= dω/dk on the basis of superposition of waves. Show that group velocity is equal to particle velocity. **[10 Marks]**			
	(OR)			
2.	(a) Prove on the basis of uncertainty principle that free electron does not exist inside the nucleus. **[10 Marks]**	20	1	L3
	(b) A spectral line of wavelength 4000 Å has a width of 8x10^{-5} Å. Evaluate the minimum time spent by the electrons in the upper energy state between the excitation and de-excitation processes. **[10 Marks]**			

3.	(a) What do you understand by normalization of wave function? A particle limited to the x axis has the wave function Ψ=ax between x=0 and x=1. Find the probability that the particle can be found between x=0.45 and x=0.55. **[12 Marks]** (b) Describe: (i) Wave function (ii) Probability Density (iii) Eigen Value (iv) Eigen Function **[8 Marks]**	20	2	L3
	(OR)			
4.	(a) For an electron in one dimensional infinite potential well of width a, discuss the solution of Schrodinger equation and hence show that the electron's energy is quantized. **[10 Marks]** (b) An electron is bound to potential well of width 0.12 nm. Find the energy values and the de Broglie wavelength in the ground state and 1st excited state. **[10 Marks]**	20	2	L3
5.	(a) Write the Fermi-Dirac distribution function. With a neat diagram explain how Fermi factor varies with temperature. **[10 Marks]** (b) Calculate the probability of an electron occupying an energy level 0.02 eV above and below the Fermi level at 400 K. **[10 Marks]**	20	3	L3
	(OR)			
6.	(a) What are majority and minority charge carriers? Explain Fermi Level in an intrinsic semiconductor. **[10 Marks]** (b) The resistivity of intrinsic germanium at 27 °C is equal to 0.47 ohm-meter. Assuming electron and hole mobilities as 0.38 and 0.18 $m^2V^{-1}s^{-1}$ respectively, calculate the intrinsic carrier density. **[10 Marks]**	20	3	L3
7.	(a) Define the bending moment of a beam and derive an expression for the bending moment of a rectangular beam. **[10 Marks]** (b) Calculate the angular twist of a wire of length 0.3 m, and radius $0.2 \times 10^{-3} m$ when a torque of $5 \times 10^{-4} Nm$ is applied. Rigidity modulus of its material is $8.0 \times 10^{10} N/m^2$. **[10 Marks]**	20	4	L4
	(OR)			
8.	(a) Derive an expression for internal field in case of one-dimensional array of atoms in dielectric solids. **[10 Marks]** (b) Define polarization and polarizability. Find the polarization produced in a dielectric medium of relative permittivity 15 in	20	4	L4

Register Number

	presence of an electric field 500 V/m. **[10 Marks]**			
9.	**(a)** Explain the principle, construction and working of a Semiconductor laser with appropriate diagram. **[10Marks]**	20	5	L
	(b) The average output power of a laser source emitting laser beam of wavelength 6328 A° is 5 mW. Find the number of photons emitted per second by laser source? **[10 Marks]**			
	(OR)			
10.	**(a)** A glass-clad fiber is made with core glass of refractive index 1.5 and the cladding is doped to give a fractional index difference of 0.0005. Find the (1) cladding index (2) critical internal reflection angle (3) external critical acceptance angle (4) numerical aperture **[10 Marks]**	20	5	I
	(b) Explain the different types of optical fibers with the necessary diagrams. **[10 Marks]**			

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CHRIST (Deemed to be University), Bengaluru – 560 029

School of Engineering and Technology

MID SEMESTER EXAMINATION – February 2024

Programme Name: B.Tech (First Year, Physics Cycle) **Max. Marks: 50**

Course Name: PHYSICS FOR ENGINEERS **Time: – 2.00Hr**

Course Code: PH232P

Course Outcomes (COs):

The students will able to:

1. Explain the basic principles of quantum mechanics and its applications to simple systems.
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- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer booklet.
- **The question paper is divided into two sections. Kindly read the instructions against each section.**
- **Constants: h=6.63 × 10^{-34} Js; charge of electron e=1.6 x 10^{-19} C; speed of light c=3 x 10^{8} m/s; mass of electron, m=9.1 x 10^{-31} kg, Boltzaman constant k_b =1.38 x 10^{-23} J/K, 1 eV=1.6 x 10^{-19} J**

SECTION A (Answer ALL questions)

Q. No	Questions	Marks	CO	RBT
1 (a)	Explain Compton Effect with necessary diagram.	4	1	3
1 (b)	Determine the change in wavelength when the photon is scattered by (i) 0°, (ii) 30°, (iii) 45°, (iv) 60°, (v) 90° and (v) 180°.	6		
2 (a)	State uncertainty principle. Write its mathematical form for different pair of variables.	5	1	3

2 (b)	In a measurement of position and velocity of an electron moving with a speed of 6×10^5 m/s, calculate the highest accuracy with which its position could be determined if the inherent error in the measurement of its velocity is 0.01% for the speed stated.	5		
3	Describe the principle, construction and working of a semiconductor diode laser with labelled energy band diagram.	10	2	3
4 (a)	Explain the different types of optical fibers with necessary labelled diagram.	6	2	3
4 (b)	If an optical fiber has glass core of refractive index 1.5 and cladding of refractive index 1.48, calculate a) numerical aperture and b) angle of acceptance if the fiber is kept in air.	4		

SECTION B (Answer any 1 question)

Q. No	Questions	Marks	CO	RBT
1	Derive time-independent one-dimensional Schrödinger's wave equation.	10	1	3
2 (a)	What is wave-function?	2	1	3
2 (b)	A particle moving in one dimension is associated with the wave function = B x^3 bounded between the region x=1 to x=2. Find the normalization constant.	8		

Revised Bloom's Taxonomy (RBT) Levels :		
L1 – Remembering	L2 – Understanding	L3 – Applying
L4 – Analyzing	L5 – Evaluating	L6 - Creating